

A PROJECT IDEA PRESENTATION ON

FlowML

Visual AI-Powered Machine
Learning Pipeline Builder

Presented by :

Team **AutoNova**

KIT's College of Engineering, Kolhapur

List of Contents

- | | |
|----------------------|------------------------|
| 1. Introduction | 5. System Architecture |
| 2. Problem Statement | 6. Tech Stack |
| 3. Objectives | 7. Use Case |
| 4. Proposed System | 8. Conclusion |

Introduction

- Machine learning enables intelligent decision-making, but building complete workflows requires technical expertise and multiple tools, making the process complex.
- Managing steps like preprocessing, training, and evaluation separately increases development time and reduces productivity.
- Beginners often struggle to understand how different stages of a machine learning pipeline connect.
- FlowML provides a visual drag-and-drop interface to automate preprocessing, model selection, and evaluation.
- This simplifies machine learning workflows, improves productivity, and makes AI accessible to more users.



Problem Statement

Challenges

- Machine learning pipelines involve multiple complex steps requiring strong technical expertise and experience.
- Beginners struggle to understand workflows, while professionals lose time switching between multiple tools.

Solution Vision

- A unified visual platform simplifies machine learning workflows and reduces technical complexity and effort.
- Users build pipelines faster with automation, improving productivity and reducing development time.

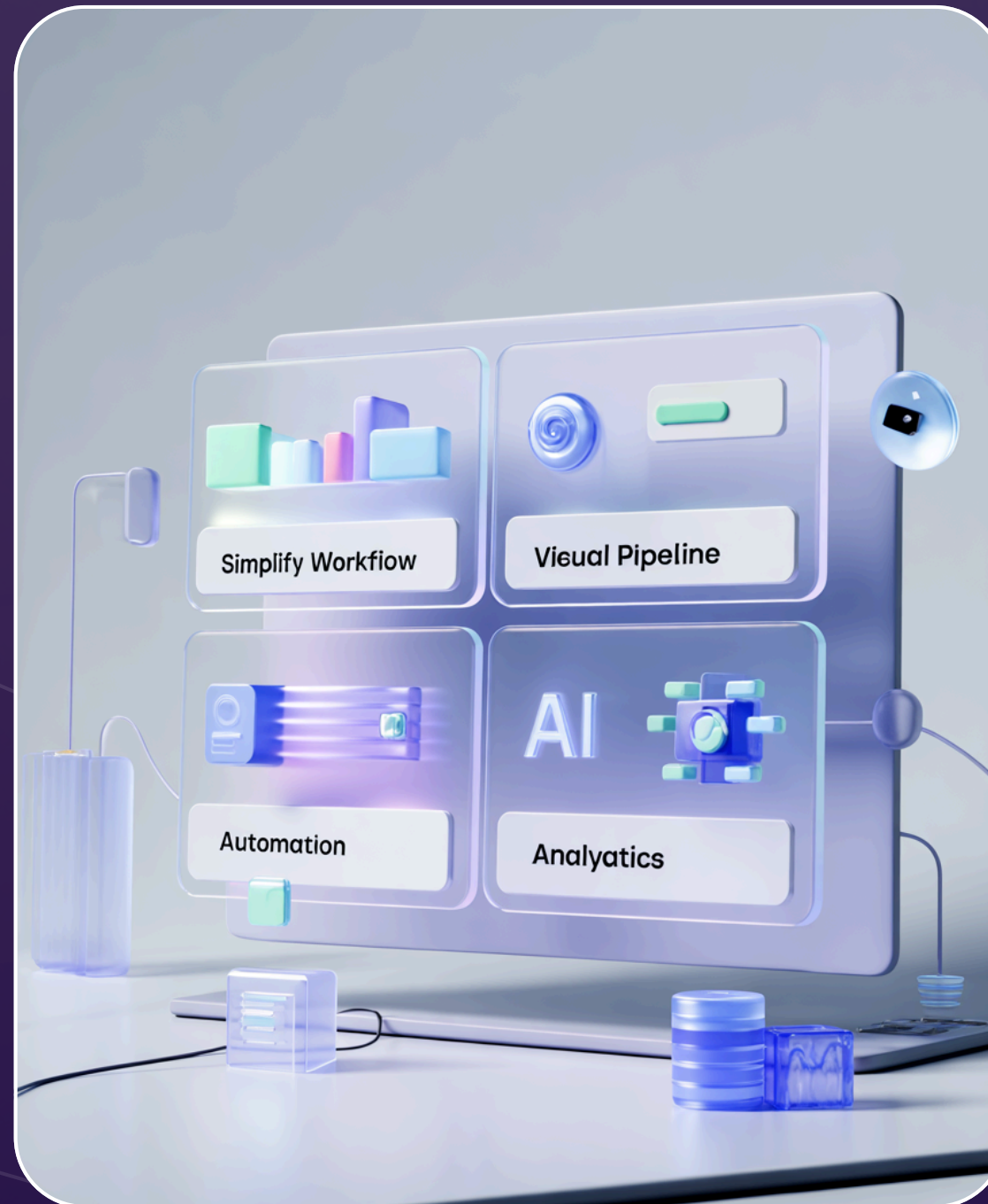
Objectives

Simplify

Simplify end-to-end machine learning workflows, improving productivity and reducing development time.

Visualize

Enable visual drag-and-drop pipeline building, simplifying workflow design and model development.



Automate

Automate preprocessing and model selection, reducing manual effort and improving efficiency.

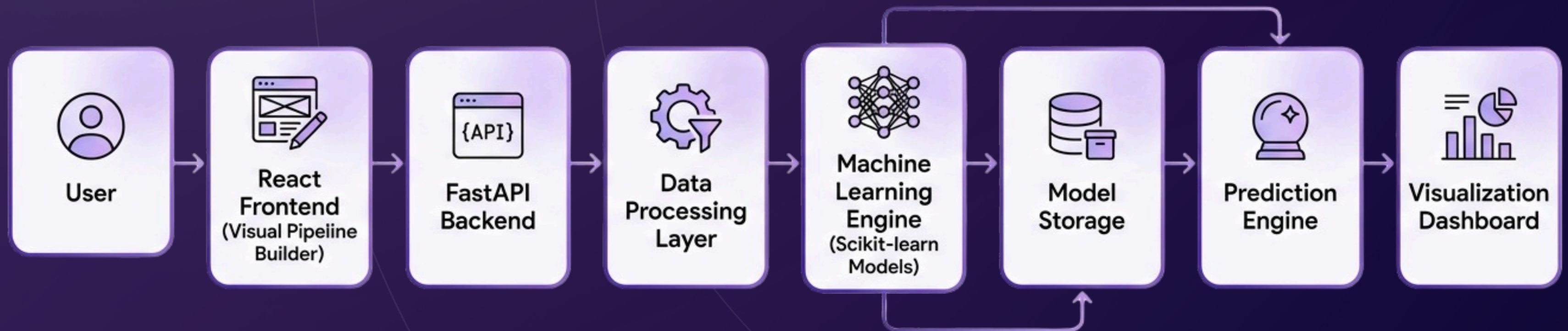
Analyze

Provide model comparison and performance insights, supporting better decision-making.

System Architecture

Architecture Components:

- React Frontend - Visual pipeline builder interface
- FastAPI Backend - Workflow execution and API handling
- Data Processing Layer - Dataset preprocessing engine
- Machine Learning Layer - Scikit-learn model training
- Visualization Layer - Charts and insights dashboard



Tech Stack

Modern tools and frameworks enabling seamless, scalable, and efficient FlowML development.



Backend Tech

1. FastAPI
2. Python
3. REST APIs
4. Async Processing
5. Modular Architecture

Frontend Tech

1. React
2. Tailwind CSS
3. React Flow
4. Framer Motion
5. Modern UI Components

Machine Learning

1. Scikit-learn
2. Pandas
3. NumPy

Use Cases

Learning

Students learn machine learning concepts through visual workflows and hands-on experimentation.

Development

Developers build machine learning applications faster using simplified pipeline creation.



Experimentation

Data scientists perform rapid experimentation and compare multiple models efficiently.

Analytics

Organizations build prediction systems for business analytics and data-driven decisions.

Conclusion

- FlowML simplifies machine learning development with a visual workflow platform that reduces technical complexity. The drag-and-drop builder allows users to create pipelines without extensive coding.
- The platform automates preprocessing, model training, and evaluation, enabling faster experimentation and improved productivity.
- FlowML also provides visual insights and prediction capabilities, making machine learning more accessible, efficient, and scalable.



Our Team

Ajinkya Walvekar

Backend & ML Integration

Vishwajeet Chandole

Frontend & UI Development

Aditya Hajare

Testing & Pipeline Design

Prathmesh Kagale

Research & Documentation

Thank You

FlowML - Visual Machine Learning Platform